

1. In my cupboard I have black socks, white socks, red socks, and Rudolph the Red Nose Reindeer socks.

How many socks do I have to take out of the cupboard to guarantee that I have a matching pair of socks?
2. Show that there will always be two pupils in the School with the same first and last initials.
3. A restaurant has five flavours of ice cream on its menu, and you can choose three different flavours for dessert. On a boarders' outing 12 pupils all order ice cream for desert. Show that at least two of them have selected the same flavours.
4. At the Remove Ball five boys and seven girls are seated around a circular table. Show that there are two girls sitting opposite each other.
5. **(Possibly already done)** Show that at any party there will always be two people who have exactly the same number of friends at the party.
6. Show that in any collection of six integers there exists at least one pair of integers whose difference is a multiple of five.
7. 15 students take a quiz, and their scores sum to 100. Show that there are at least two students that have the same score.
8.
 - a. There are 30 chairs in a row. What is the maximum number of people that can be seated without having a consecutive set of four chairs occupied?
 - b. What if the chairs are in three rows of 10?
9. Show that if 1000 different numbers are chosen from the $1, 2, \dots, 1997$ then there must be at least one pair of numbers that sums to 1998.
10. A student revising for an important exam does at least one past paper each day for the seven weeks leading up to the exam. Because of time constraints, the pupil cannot do more than 11 past papers in any one week. Show that there must be some period of consecutive days in which the pupil has done exactly 20 past papers.
11. 101 positive integers whose sum is 300 are placed in a circle. Prove that is it possible to choose some adjacent numbers from these numbers whose sum is equal to 200.
12. What is the maximum size of subset of $\{1, 2, \dots, 2n\}$ such that no element of the subset is a multiple of any other element of the subset?
13. The edges of the complete graph on n vertices are all coloured either red or blue. Show that provided that $n \geq 6$ there is at least one red or blue triangle.
14. Every lattice point on the plane is coloured either red, green, or blue. Prove that there exists a rectangle in the plane such that all four of its vertices are the same colour.